FDS EXPERIMENT-8

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Conduct Z test for the given sample data using Python.

```
import numpy as np
```

```
values = np.array(list(map(int,input("Enter the values of the array : ").split(" "))))
```

```
Enter the values of the array : 112 118 115 120 119 114 117 122 116 121
```

```
sample_mean = 108
alpha = 0.05
critical_value = 1.96
```

```
population_mean = values.mean()
population_std = values.std()
n = len(values)
```

```
print("Population mean :",population_mean)
print("Sample mean :",sample_mean)
print("Population standard deviation :",population_std)
print("Critical value :",critical_value)
print("Alpha value :",alpha)
```

```
Population mean : 117.4
Sample mean : 108
Population standard deviation : 3.0397368307141326
Critical value : 1.96
Alpha value : 0.05
```

```
#Compute Z value
Z_calculated = (sample_mean - population_mean)/(population_std / (n**0.5))
print("Z value :",Z_calculated)
```

```
Z value : -9.778941948273635
```

```
#Compare Z value with critical value

if(abs(Z_calculated) > critical_value):
  print("Conclusin : Reject the null Hypothesis. The sample mean is significantly different.....")

else:
  print("Conclusion : Accept the null Hypothesis.......")
```

```
Conclusin : Reject the null Hypothesis. The sample mean is significantly different.....
```